

A Fully Parallelized Scheme of Constructing Independent Spanning Trees on Möbius Cubes *

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Abstract

A set of spanning trees in a graph is said to be independent (ISTs for short) if all the trees are rooted at the same node r and for any other node $v(\neq r)$, the paths from v to r in any two trees are node-disjoint except the two end nodes v and r . It was conjectured that for any n -connected graph there exist n ISTs rooted at an arbitrary node. Recently, Cheng et al. [6] and [3] respectively proposed algorithms to run in $\mathcal{O}(N \log N)$ time and $\mathcal{O}(N)$ time for constructing n ISTs rooted at an arbitrary node on an n -dimensional Möbius cube MQ_n , where $N = 2^n$ is the number of nodes in MQ_n . However, both algorithms are executed in a recursive fashion and thus are hard to be parallelized. Although the algorithm in [3] can simultaneously construct n ISTs, it is not fully parallelized for the construction of each spanning tree. In this paper, we present a non-recursive and fully parallelized approach to construct n ISTs rooted at an arbitrary node of MQ_n in $\mathcal{O}(\log N)$ time.

Keyword: independent spanning trees; interconnection networks; Möbius cubes;

1 Introduction

Constructing multiple spanning trees in networks have been studied from not only the theoretical point of view but also some practical applications such as fault-tolerant broadcasting [1, 14] and secure message distribution [1, 22, 29]. Let G be a graph with node set $V(G)$ and edge set $E(G)$, respectively. Two spanning trees in a graph G are said to be *independent* if they are rooted at the

same node r such that, for each node $v(\neq r)$ in G , the two different paths from v to r , one path in each tree, are internally node-disjoint. A set of spanning trees of G is called *independent spanning trees* (ISTs for short) if they are pairwise independent.

A graph G is *k-connected* if $|V(G)| > k$ and $G - F$ is connected for every subset $F \subseteq V(G)$ with $|F| < k$, where $G - F$ denotes the graph obtained from G by removing F . Zehavi and Itai [36] proposed the following conjecture: If G is a k -connected graph and r is an arbitrary node of G , then G has k ISTs rooted at r . From then on, this conjecture has been shown to be true for k -connected graphs with $k \leq 4$ (see [14], [7, 36] and [9] for $k = 2, 3, 4$, respectively) and is still open for $k \geq 5$. In particular, this conjecture has been confirmed for several restricted classes of graphs, e.g., graphs related to planarity [12, 13, 19, 20], graphs defined by Cartesian product [2, 21, 23, 24, 27, 31, 35], variations of hypercubes [4–6, 18, 25, 26, 29], special Cayley graphs [16, 17, 22, 30, 33, 34], and chordal ring [15, 32].

The family of Möbius cubes was first introduced by Cull and Larson [8] as a variation of hypercubes to achieving some improvements of structure properties, even though they possess of the property of asymmetry. One advantage of the Möbius cube is that the diameter is only about one half of the diameter of the equivalent hypercube. Xu and Deng [28] showed that the wide diameter of the Möbius cube is at most $\lceil (n+2)/2 \rceil + 2$. Since it was pointed out in [8] that the n -dimensional Möbius cube MQ_n has the connectivity n , Cheng et al. [6] and [3] respectively proposed recursive algorithms to construct n ISTs rooted at an arbitrary node in MQ_n . Let $N = 2^n$. The construction in [6] can be run in $\mathcal{O}(N \log N)$ time. Although the algorithm in [3] can simultaneously construct n ISTs in paral-

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lel and is with time complexity $\mathcal{O}(N)$, it is not fully parallelized for the construction of each spanning tree (in fact, the constructing scheme looks like a recursive construction of a binomial-like tree). For more properties of Möbius cubes, the reader can refer to [8, 10, 11, 28].

In this paper, we present a non-recursive and fully parallelized approach for constructing n ISTs rooted at an arbitrary node in MQ_n . The rest of this paper is organized as follows. Section 2 formally gives the definition of Möbius cubes and provides some useful terminologies and notations. Section 3 presents our algorithm for constructing n ISTs in MQ_n . Section 4 proves the correctness of the algorithm. The final section contains our concluding remarks.

2 Preliminary

The n -dimensional Möbius cube, denoted by MQ_n , is a variant of the n -dimensional hypercube, where the *label* or *address* of each node is represented by a unique binary string of length n . Two types of n -dimensional Möbius cube include the 0-type n -dimensional Möbius cube ($0-MQ_n$ for short) and 1-type n -dimensional Möbius cube ($1-MQ_n$ for short), which are defined recursively as follows:

Definition 1. [10] Both $0-MQ_1$ and $1-MQ_1$ are the complete graph on two nodes labeled by 0 and 1. For $n \geq 2$, both $0-MQ_n$ and $1-MQ_n$ are composed of two $(n-1)$ -dimensional Möbius subcubes where one, denoted by MQ_{n-1}^0 , is a $0-MQ_{n-1}$ with a common prefix 0 in every node and the other, denoted by MQ_{n-1}^1 , is a $1-MQ_{n-1}$ with a common prefix 1 in every node, such that two nodes $x = x_{n-1}x_{n-2}\cdots x_0 \in V(MQ_{n-1}^0)$ and $y = y_{n-1}y_{n-2}\cdots y_0 \in V(MQ_{n-1}^1)$ are adjacent in $0-MQ_n$ (respectively, in $1-MQ_n$) if and only if $x_{n-1} = \bar{y}_{n-1} = 0$ and $x_i = y_i$ for all $i = 0, 1, \dots, n-2$ (respectively, $x_{n-1} = \bar{y}_{n-1} = 0$ and $x_i = \bar{y}_i$ for all $i = 0, 1, \dots, n-2$).

Figure 1 illustrates the two types of Möbius cubes $0-MQ_4$ and $1-MQ_4$, respectively. In the rest of this paper, for convenience of discussion, we say MQ_n to mean either $0-MQ_n$ or $1-MQ_n$ if there are no ambiguities, whereas we denote MQ_n^0 and MQ_n^1 to distinguish $0-MQ_n$ and $1-MQ_n$, respectively. That is, we regard the label of every node in the $0-MQ_n$ (respectively, $1-MQ_n$) with an additionally preceding bit 0 (respectively, bit 1). Accordingly, the Möbius cube can also be equivalently defined by the following non-recursive fashion:

Definition 2. [6] For all $n \geq 1$, two nodes $x = x_{n-1}x_{n-2}\cdots x_0$ and $y = y_{n-1}y_{n-2}\cdots y_0$ are joined by an edge in MQ_n if and only if there exists an integer $i \geq 0$ such that

- (1) $x_{n-1}x_{n-2}\cdots x_{i+1} = y_{n-1}y_{n-2}\cdots y_{i+1}$,
- (2) $x_i \neq y_i$,
- (3) $x_{i-1}x_{i-2}\cdots x_0 = y_{i-1}y_{i-2}\cdots y_0$ if $x_{i+1} = 0$, and
- (4) $x_{i-1}x_{i-2}\cdots x_0 = \bar{y}_{i-1}\bar{y}_{i-2}\cdots \bar{y}_0$ if $x_{i+1} = 1$.

In the above definition, if both conditions (1) and (2) hold, then we say that x and y have the *leftmost differing bit at position i* . In this case, x and y are said to be the i -neighbor to each other, and for notational convenience we write $y = N_i(x)$ or $x = N_i(y)$. For example, if we consider the node $x = 011011$ in $0-MQ_6$ (respectively, in $1-MQ_6$), then $N_0(x) = 011010$, $N_1(x) = 011001$, $N_2(x) = 011100$, $N_3(x) = 010100$, $N_4(x) = 001011$, and $N_5(x) = 111011$ (respectively, $N_5(x) = 100100$).

In this paper, we also use the following notation. Two paths P and Q joining two distinct nodes x and y are *internally node-disjoint*, denoted by $P \parallel Q$, if $V(P) \cap V(Q) = \{x, y\}$. Let T be a spanning tree rooted at node r of MQ_n . The parent of a node $x (\neq r)$ in T is denoted by $\text{PARENT}(T, x)$. For $x, y \in V(T)$, the unique path from x to y is denoted by $T[x, y]$. Hence, two spanning trees T and T' with the same root r are ISTs if and only if $T[x, r] \parallel T'[x, r]$ for every node $x \in V(T) \setminus \{r\}$.

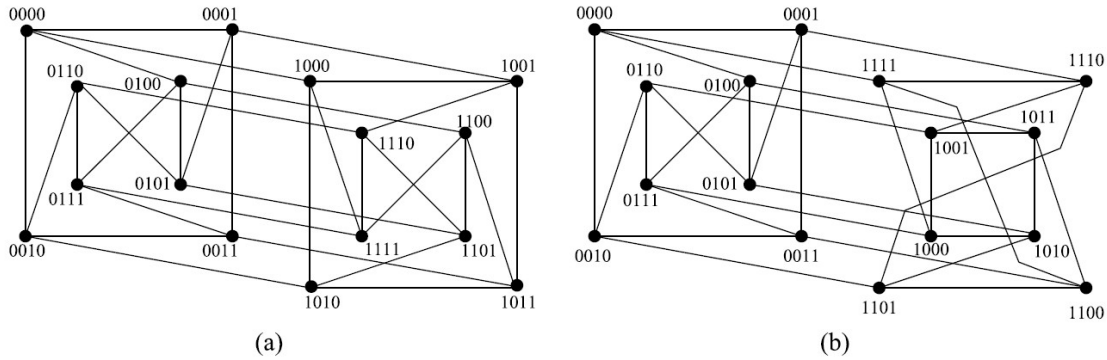


Figure 1: (a) $0-MQ_4$; (b) $1-MQ_4$

3 Constructing ISTs on Möbius cubes in parallel

Let $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$. Since MQ_n is n -connected and we would like to construct n ISTs, the root in each spanning tree must have a unique child. We arbitrarily choose a node $r \in V(MQ_n)$ as the root. For $i \in \mathbb{Z}_n$, we denote T_i as a tree such that r takes its i -neighbor as the unique child. Let $N_i(r) = c_{n-1}c_{n-2} \cdots c_0$. For each node $x = x_{n-1}x_{n-2} \cdots x_0 \in V(MQ_n) \setminus \{r\}$ and $i \in \mathbb{Z}_n$, we define

$$D_i(x) = \{j \in \mathbb{Z}_n \setminus \mathbb{Z}_{i+1} : c_{j+1} = 1 \text{ and } x_j \neq c_j\}$$

and

$$S_i(x) = \{j \in \mathbb{Z}_n \setminus \mathbb{Z}_{i+1} : c_{j+1} = 1 \text{ and } x_j = c_j\}.$$

Also, a node called the *surrenal* of $N_i(r)$ with respect to x is defined as follows:

$$\begin{aligned} \bar{N}_i(r, x) &= c'_{n-1}c'_{n-2} \cdots c'_0 \\ &= \begin{cases} c_{n-1}c_{n-2} \cdots c_0 & \text{if } D_i(x) = \emptyset \text{ or} \\ & \min D_i(x) > \min S_i(x); \\ c_{n-1}c_{n-2} \cdots c_{i+1}\bar{c}_i\bar{c}_{i-1} \cdots \bar{c}_0 & \text{otherwise.} \end{cases} \end{aligned}$$

Let $I_i(x) = \{j \in \mathbb{Z}_n : x_j \neq c'_j\}$. Furthermore, we define the following function:

$$\text{NEXT}(i, x) = \begin{cases} i & \text{if } I_i(x) = \emptyset; \\ \max I_i(x) & \text{if } I_i(x) \neq \emptyset \text{ and } i < \min I_i(x); \\ \max\{j \in I_i(x) : j \leq i\} & \text{otherwise.} \end{cases}$$

That is, we regard $I_i(x)$ as a cyclic ordered set in decreasing order. If $I_i(x) = \emptyset$ or $i \in I_i(x)$, the function outputs i ; otherwise, the function outputs the next element in the cyclic order of $I_i(x)$ with respect to i .

It is clear that, for each node $x \in V(MQ_n) \setminus \{r\}$, finding $D_i(x)$, $S_i(x)$, $\bar{N}_i(r, x)$, $I_i(x)$ and $\text{NEXT}(i, x)$ can be done in $\mathcal{O}(n)$ time provided i is given. In what follows, we present a fully parallelized algorithm for constructing n spanning trees with an arbitrary node $r = r_{n-1}r_{n-2} \cdots r_0$ as their common root in MQ_n . For each node $x = x_{n-1}x_{n-2} \cdots x_0 \in V(MQ_n) \setminus \{r\}$, the construction can be carried out by describing the parent of x in each spanning tree T_i . Figure 2 shows the algorithm.

Example 1. Consider $0-MQ_5$ and suppose we choose $r = 10111_2 = 23$ as the common root. For conciseness, we represent a node in $0-MQ_5$ by decimal. We describe how the algorithm CONSTRUCTING-ISTs constructs T_0 in $0-MQ_5$ as

Algorithm CONSTRUCTING-ISTs

Input: All nodes of MQ_n and the common root

$$r = r_{n-1}r_{n-2} \cdots r_0.$$

Output: n ISTs $T_0, T_1, \dots, T_{n-1}$ root at r .

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1:  for  $i = 0$  to  $n-1$  do in parallel
    /* construct  $T_i$  simultaneously */
2:    for each node  $x$  in  $MQ_n$  do in parallel
    /* generate parent of each node  $x$ 
       simultaneously */
3:       $j = \text{NEXT}(i, x)$ 
4:       $\text{PARENT}(T_i, x) = N_j(x)$ 

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Figure 2: Algorithm for constructing n spanning trees in MQ_n .

follows. Clearly, the 0-neighbor of r is $N_0(23) = 10110_2 = 22$. Let $x = 11010_2 = 26$ and $i = 0$. Clearly, $\mathbb{Z}_5 \setminus \mathbb{Z}_{i+1} = \{1, 2, 3, 4\}$. Since $c_2 = c_4 = 1$, $x_1 = c_1$ and $x_3 \neq c_3$, we have $D_0(x) = \{3\}$ and $S_0(x) = \{1\}$. Since $\min D_0(x) > \min S_0(x)$, we have $\bar{N}_0(r, x) = 10110_2$, and thus $I_0(x) = \{3, 2\}$ and $j = \text{NEXT}(0, x) = 3$. It follows that $\text{PARENT}(T_0, x) = N_3(x) = 10101_2 = 21$. Let $y = 21$. Since $c_2 = c_4 = 1$, $y_1 \neq c_1$ and $y_3 = c_3$, we have $D_0(y) = \{1\}$ and $S_0(y) = \{3\}$. Since $\min D_0(y) < \min S_0(y)$, we have $\bar{N}_0(r, y) = 10111_2$, and thus $I_0(y) = \{1\}$ and $j = \text{NEXT}(0, y) = 1$. It follows that $\text{PARENT}(T_0, y) = N_1(y) = 10110_2 = 22$. Let $z = 22$. Since $c_2 = c_4 = 1$, $z_1 = c_1$ and $z_3 = c_3$, we have $D_0(z) = \emptyset$ and $S_0(z) = \{1, 3\}$. It further implies $\bar{N}_0(r, z) = 10110_2$, and thus $I_0(z) = \emptyset$ and $j = \text{NEXT}(0, z) = 0$. It follows that $\text{PARENT}(T_0, z) = N_0(z) = 10111_2 = r$. Due to the limitation of space, we only summarize details of the construction of T_0 for $0-MQ_5$ in Table 1, and provide all constructing results in Figure 3. For convenience, we adopt the notation $x \xrightarrow{j} y$ to mean that $y = N_j(x)$ in MQ_n . For instance, we have $T_0[26, 23] : 26 \xrightarrow{3} 21 \xrightarrow{1} 22 \xrightarrow{0} 23$ in Figure 3.

4 Correctness

In this section, we will show the validity of the algorithm. For two ordered sets A and B , we write $A \prec_{\text{LEX}} B$ to mean that A precedes B in lexicographic order. Firstly, we prove the reachability between every node $x (\neq r)$ and the root r in T_i , thereby proving the existence of a unique path from x to the root in the tree.

Lemma 1. *Let $r \in V(MQ_n)$ be an arbitrary node. The constructions of T_i for all $i \in \mathbb{Z}_n$ are spanning trees rooted at r .*

Proof. From CONSTRUCTING-ISTs, since every

Table 1: The parent of nodes $x \in V(MQ_5^0) \setminus \{23\}$ in T_0 with root $r = 10111_2 = 23$. $i = 0, N_0(23) = 10110_2 = 22$

x	binary string	$\min D_0(x)$	$\min S_0(x)$	$\bar{N}_0(r, x)$	$I_0(x)$	$j = \text{NEXT}(0, x)$	$\text{PARENT}(T_0, x)$
0	00000	1	3	10111	$\{4, 2, 1, 0\}$	0	$= N_0(00000) = 00001 = 1$
1	00001	1	3	10111	$\{4, 2, 1\}$	4	$= N_4(00001) = 10001 = 17$
2	00010	-	1	10110	$\{4, 2\}$	4	$= N_4(00010) = 10010 = 18$
3	00011	-	1	10110	$\{4, 2, 0\}$	0	$= N_0(00011) = 00010 = 2$
4	00100	1	3	10111	$\{4, 1, 0\}$	0	$= N_0(00100) = 00101 = 5$
5	00101	1	3	10111	$\{4, 1\}$	4	$= N_4(00101) = 10101 = 21$
6	00110	-	1	10110	$\{4\}$	4	$= N_4(00110) = 10110 = 22$
7	00111	-	1	10110	$\{4, 0\}$	0	$= N_0(00111) = 00110 = 6$
8	01000	1	-	10111	$\{4, 3, 2, 1, 0\}$	0	$= N_0(01000) = 01001 = 9$
9	01001	1	-	10111	$\{4, 3, 2, 1\}$	4	$= N_4(01001) = 11001 = 25$
10	01010	3	1	10110	$\{4, 3, 2\}$	4	$= N_4(01010) = 11010 = 26$
11	01011	3	1	10110	$\{4, 3, 2, 0\}$	0	$= N_0(01011) = 01010 = 10$
12	01100	1	-	10111	$\{4, 3, 1, 0\}$	0	$= N_0(01100) = 01101 = 13$
13	01101	1	-	10111	$\{4, 3, 1\}$	4	$= N_4(01101) = 11101 = 29$
14	01110	3	1	10110	$\{4, 3\}$	4	$= N_4(01110) = 11110 = 30$
15	01111	3	1	10110	$\{4, 3, 0\}$	0	$= N_0(01111) = 01110 = 14$
16	10000	1	3	10111	$\{2, 1, 0\}$	0	$= N_0(10000) = 10001 = 17$
17	10001	1	3	10111	$\{2, 1\}$	2	$= N_2(10001) = 10101 = 21$
18	10010	-	1	10110	$\{2\}$	2	$= N_2(10010) = 10110 = 22$
19	10011	-	1	10110	$\{2, 0\}$	0	$= N_0(10011) = 10010 = 18$
20	10100	1	3	10111	$\{1, 0\}$	0	$= N_0(10100) = 10101 = 21$
21	10101	1	3	10111	$\{1\}$	1	$= N_1(10101) = 10110 = 22$
22	10110	-	1	10110	$\emptyset$	0	$= N_0(10110) = 10111 = 23$
23	-	-	-	-	-	-	(root)
24	11000	1	-	10111	$\{3, 2, 1, 0\}$	0	$= N_0(11000) = 11001 = 25$
25	11001	1	-	10111	$\{3, 2, 1\}$	3	$= N_3(11001) = 10110 = 22$
26	11010	3	1	10110	$\{3, 2\}$	3	$= N_3(11010) = 10101 = 21$
27	11011	3	1	10110	$\{3, 2, 0\}$	0	$= N_0(11011) = 11010 = 26$
28	11100	1	-	10111	$\{3, 1, 0\}$	0	$= N_0(11100) = 11101 = 29$
29	11101	1	-	10111	$\{3, 1\}$	3	$= N_3(11101) = 10010 = 18$
30	11110	3	1	10110	$\{3\}$	3	$= N_3(11110) = 10001 = 17$
31	11111	3	1	10110	$\{3, 0\}$	0	$= N_0(11111) = 11110 = 30$

node $v \in V(MQ_n)$ must be contained in T_i , it follows that T_i is a spanning subgraph of MQ_n . Let $x = x_{n-1}x_{n-2} \cdots x_0$ be any node of MQ_n . In the following, we show that $T_i[x, r]$ is the unique path connecting x and r in T_i . Clearly, if $I_i(x) = \emptyset$, then $x = \bar{N}_i(r, x) = N_i(r)$ and $\text{NEXT}(i, x) = i$. Thus, $T_i[x, r] : x \xrightarrow{i} r$ is the desired path connecting x and r in T_i .

Next, we suppose that $I_i(x) = \{j_{p-1}, j_{p-2}, \dots, j_0\}$ is nonempty and is treated as an ordered set such that $j_{p-1} > j_{p-2} > \cdots > j_0$. Clearly, $1 \leq p \leq n$. By definition, $x_j \neq c'_j$ for $j \in I_i(x)$ and $x_j = c'_j$ for $j \in \mathbb{Z}_n \setminus I_i(x)$. There are two scenarios as follows:

Case 1: $i \notin I_i(x)$ (i.e., $x_i = c'_i$). Let $j_k = \text{NEXT}(i, x)$, where $0 \leq k \leq p-1$. Since $I_i(x) \neq \emptyset$, we assume that $y(\neq r) = y_{n-1}y_{n-2} \cdots y_0$ is the parent of x in T_i . That is, $y = \text{PARENT}(T_i, x) = N_{j_k}(x)$. Clearly, $y_{j_k} = \bar{x}_{j_k}$. Consider the following two cases:

Case 1.1: $x_{j_k+1} = 0$. In this case, we have $y_j = x_j$ for $j_k > j \geq 0$. Since $D_i(y) = D_i(x)$ and $S_i(y) = S_i(x)$, it implies $\bar{N}_i(r, y) = \bar{N}_i(r, x)$, and thus $I_i(y) = I_i(x) \setminus \{j_k\}$.

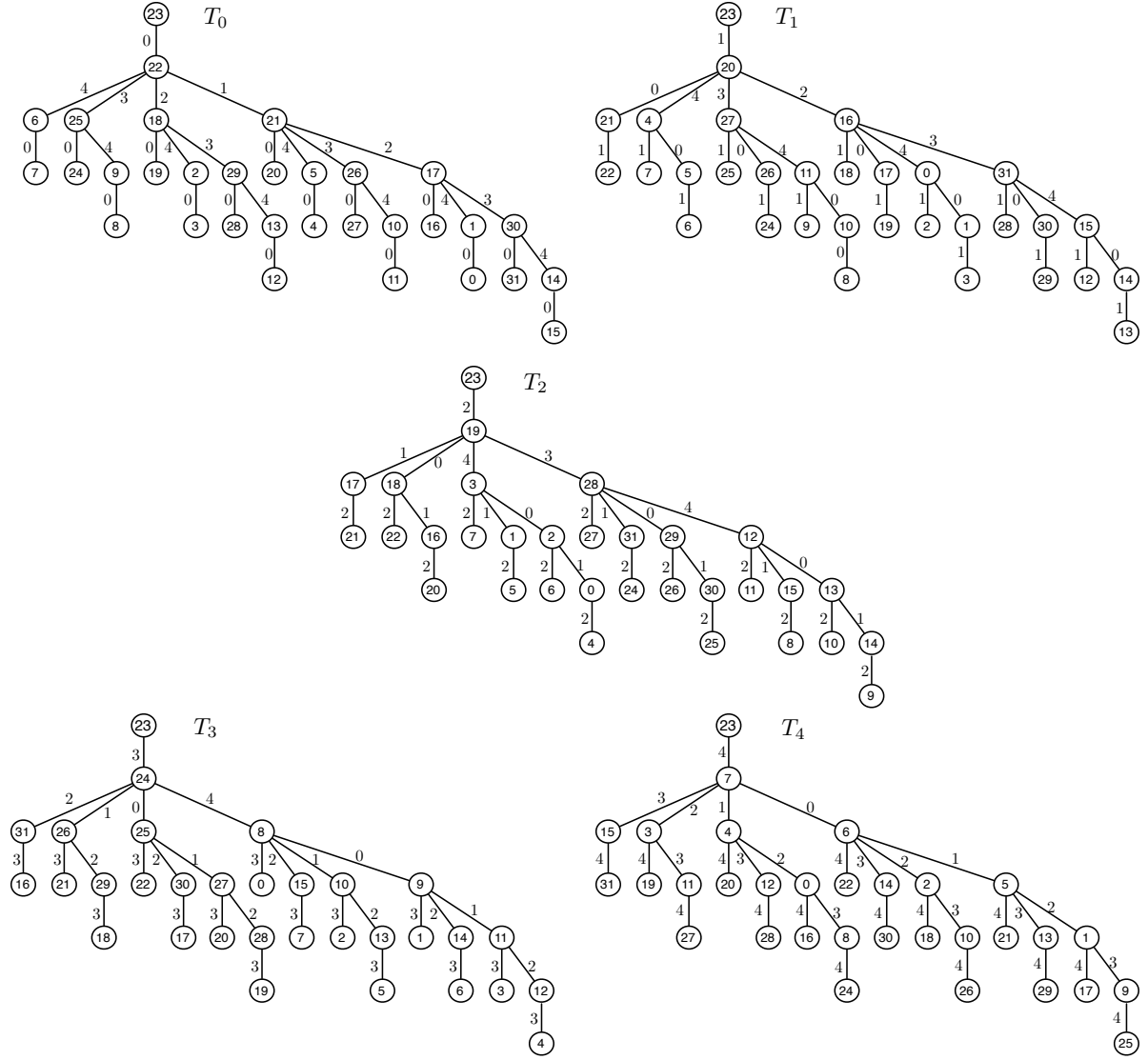
Case 1.2: $x_{j_k+1} = 1$. In this case, we have $y_j = \bar{x}_j$ for $j_k > j \geq 0$. Since $D_i(y), S_i(y) \subseteq \mathbb{Z}_n \setminus \mathbb{Z}_{i+1}$, if $i > j_k$ then $D_i(y) = D_i(x)$ and $S_i(y) = S_i(x)$. It further implies $\bar{N}_i(r, y) = \bar{N}_i(r, x)$, and thus $I_i(y) =$

$(I_i(x) \cup \mathbb{Z}_{j_k}) \setminus \{j_k, j_{k-1}, \dots, j_0\}$. On the other hand, we have $j_k = j_{p-1} \geq j_0 > i$. Obviously, if $j \in I_i(x)$ then $y_j = \bar{x}_j = c'_j$. Moreover, we have $y_j = x_j = c'_j$ for $j \in \mathbb{Z}_n \setminus \mathbb{Z}_{j_k+1}$ and $y_j = \bar{x}_j \neq c'_j$ for $j \in \mathbb{Z}_{j_k+1} \setminus I_i(x)$. Thus, if $\bar{N}_i(r, y) = \bar{N}_i(r, x)$, then $I_i(y) = \mathbb{Z}_{j_k+1} \setminus I_i(x)$; otherwise, $I_i(y) = (\mathbb{Z}_{j_k+1} \setminus \mathbb{Z}_{i+1}) \setminus I_i(x)$.

From above, we can determine $I_i(y)$. In particular, we show that $I_i(y) \prec_{\text{LEX}} I_i(x)$ and $j_k \notin I_i(y)$. By a similar argument, if $I_i(y) \neq \emptyset$, let $z = \text{PARENT}(T_i, y) = N_{j_\ell}(y)$ be the parent of y in T_i , where $j_\ell = \text{NEXT}(i, y)$. Again, we can determine $I_i(z)$ and show that $I_i(z) \prec_{\text{LEX}} I_i(y)$ and $j_\ell \notin I_i(z)$. By this way, we find a sequence of nodes $y, z, \dots, w$ in T_i such that $I_i(w) = \emptyset$, and thus $w = N_i(r)$. Recall that we have already constructed $T_i[w, r] = w \xrightarrow{i} r$ for connecting $N_i(r)$ and r in T_i . Therefore, we obtain the following unique path that connects x and r in T_i :

$$T_i[x, r] : x \xrightarrow{j_k} y \xrightarrow{j_\ell} z \xrightarrow{j_m} \cdots \xrightarrow{j_z} w \xrightarrow{i} r.$$

Case 2: $i \in I_i(x)$ (i.e., $x_i \neq c'_i$). Suppose $i = j_k$ for some $k \in \{0, 1, \dots, p-1\}$. In this case, we have $\text{NEXT}(i, x) = i$. Let $y = \text{PARENT}(T_i, x) = N_i(x)$. Clearly, $y_i = \bar{x}_i = c'_i$. Since $D_i(y) = D_i(x)$ and $S_i(y) = S_i(x)$, it implies $\bar{N}_i(r, y) = \bar{N}_i(r, x)$, and thus $i \notin I_i(y)$. This shows that the current status for y is in the situation of Case 1. Let $P = T_i[y, r]$

Figure 3: Five ISTs of $0\text{-}MQ_5$.

be the path connecting y and r in T_i . Therefore, we obtain the unique path $T_i[x, r]$ by concatenating $x \xrightarrow{i} y$ and P . $\square$

According to the proof of Lemma 1, we have the following properties.

Corollary 2. For $i \in \mathbb{Z}_n$, let $T_i[x, r] : v_0(=x) \xrightarrow{j_1} v_1 \xrightarrow{j_2} \dots \xrightarrow{j_k} v_k \xrightarrow{i} r$ be a path constructed from Lemma 1. Then, the following statements hold:

- (1) $\emptyset = I_i(v_k) \prec_{\text{LEX}} I_i(v_{k-1}) \prec_{\text{LEX}} \dots \prec_{\text{LEX}} I_i(v_0)$.
- (2) For $1 \leq \ell < m \leq k$, $j_\ell \notin I_i(v_m)$ (i.e., $j_\ell \neq j_m$).
- (3) For $2 \leq \ell \leq k$, $j_\ell \neq i$. In particular, it is possible $j_1 = i$.

For instance, if we consider the path $T_0[15, 23] : 15 \xrightarrow{0} 14 \xrightarrow{4} 30 \xrightarrow{3} 17 \xrightarrow{2} 21 \xrightarrow{1} 22 \xrightarrow{0} 23$ in Figure 3, we can verify from Table 1 as follows:

$(I_0(22) = \emptyset) \prec_{\text{LEX}} (I_0(21) = \{1\}) \prec_{\text{LEX}} (I_0(17) = \{2, 1\}) \prec_{\text{LEX}} (I_0(30) = \{3\}) \prec_{\text{LEX}} (I_0(14) = \{4, 3\}) \prec_{\text{LEX}} (I_0(15) = \{4, 3, 0\})$. Let $\text{HEIGHT}(T)$ denote the height of a tree T . Since $|I_i(x)| \leq n$ for every node $x \in V(MQ_n)$, the following result can be obtained from Corollary 2 directly.

Corollary 3. For $i \in \mathbb{Z}_n$, $\text{HEIGHT}(T_i) \leq n + 1$.

Lemma 4. The spanning trees constructed from CONSTRUCTING-ISTs are independent.

Proof. We prove the lemma by contradiction. Suppose that the lemma is false. That is, there exist two integers $i, j \in \mathbb{Z}_n$ and a node $x \in V(MQ_n) \setminus \{r\}$ such that the following two paths constructed in Lemma 1 satisfy $\{x, r\} \subsetneq P \cap Q$:

$$P = T_i[x, r] : u_0(=x) \xrightarrow{j_0} u_1 \xrightarrow{j_1} u_2 \xrightarrow{j_2} \dots \xrightarrow{j_{k-1}} u_k \xrightarrow{i} r$$

and

$$Q = T_j[x, r] : v_0(=x) \xrightarrow{\ell_0} v_1 \xrightarrow{\ell_1} v_2 \xrightarrow{\ell_2} \dots \xrightarrow{\ell_{m-1}} v_m \xrightarrow{j} r.$$

Suppose that $u_p = v_q$ for $1 \leq p < k$ and $1 \leq q < m$. Let $A = \{j_p, j_{p+1}, \dots, j_{k-1}, i\}$ and $B = \{\ell_q, \ell_{q+1}, \dots, \ell_{m-1}, j\}$. Since $i \neq j$, by Corollary 2 we have $A \neq B$. Let $d = \max((A \cup B) \setminus (A \cap B))$. This implies that the d th bit of u_p is different from that of v_q , which leads to a contradiction. $\square$

According to Lemmas 1 and 4, we have the following theorem.

Theorem 5. *Let $N = 2^n$ and $r \in V(MQ_n)$ be an arbitrary node. Algorithm CONSTRUCTING-ISTs can correctly construct n ISTs rooted at r in $\mathcal{O}(N \log N)$ time. In particular, the algorithm can be parallelized on MQ_n by using N processors to run in $\mathcal{O}(\log N)$ time.*

5 Concluding remarks

In this paper, we provide a non-recursive and fully parallelized approach for constructing n ISTs rooted at an arbitrary node of the n -dimensional Möbius cube MQ_n in $\mathcal{O}(\log N)$ time, where $N = 2^n$ is the number of nodes in MQ_n . Indeed, all ISTs constructed in here are isomorphic to those in [6] and [3], which have height $n + 1$. To the best of our knowledge, for class of networks without node-symmetry, the present paper is the first to employ the fully parallelized approach for constructing ISTs. In particular, the properties of paths described in Corollary 2 make the proof of independency to become easier than ever before.

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